Age- and sex-standardised incidence of diabetes mellitus in Peru from 2016 to 2023

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Key words

Diabetes, epidemiology, disease burden, Peru

1. Introduction

The incidence of diabetes mellitus (DM) has increased significantly in recent decades with an age-standardised rate rising from 234 per 100,000 people in 1990 to 285 per 100,000 in 2017[1]. According to estimates by the International Diabetes Federation, the number of people living with DM in South and Central America is projected to increase to 52 million by 2050[2].

Findings from a Peruvian study showed a cumulative incidence of DM of 7.2% and an incidence rate of 19.5 (95% confidence interval [CI] 13.9 - 28.3) new cases per 1,000 person-years. These results indicate that the incidence in Peru is among the highest reported worldwide[3]. However, reports on DM incidence in Peru remain limited, and the few available studies have been conducted primarily in urban populations.

The objective of this study was to estimate age-specific, sex-specific, and age-standardised incidence rates of diabetes in Peru between 2016 and 2023 by implementing the illness–death model.

2. Subjects, Materials and Methods

2.1 Estimation of input prevalence

An observational study was conducted using cross-sectional data from the Demographic and Family Health Surveys (ENDES) for the period 2018–2023. Based on these surveys, prevalent cases of DM at the national level in Peru were identified. The ENDES survey is conducted annually and is representative of the entire Peruvian territory, providing information on demographic characteristics, health indicators across the life course, communicable and non-communicable diseases, and related determinants [4]?.

The target population comprises households and their resident members, with data collected by trained personnel who visit the selected households. The instruments for data collection include interviews and laboratory measurements[4]. Prevalent cases of DM were identified with the questions: “Has a doctor ever diagnosed you with diabetes or ‘high sugar’ in your blood?”. This question offers the response options: “yes”, “no”, and “don’t know/don’t remember”[4]. For the analyses, only participants who respond “yes” or “no” will be included, excluding those who answer “don’t know/don’t remember”.

Subsequently, age-specific prevalence of DM was estimated by sex for the years 2016-2023. Age groups were categorized in five-year intervals from 15-19 years up to 80+ years, with 95% CI calculated using logit transformation. Additionally, data from the United Nations World Population Prospects will be used to obtain mortality rates in the general population for the years 2016-2023.

2.2 Estimating the incidence rate of diabetes

To estimate incidence using the approach described below, information on the mortality rate ratio (MRR) comparing people with and without DM is required. As no studies to date have estimated the MRR for Peru, several scenarios were modeled, varying the MRR within a plausible range as done in previous studies[5]. Both the ENDES survey data and the United Nations mortality data are publicly available and can be accessed here (insert links here).

To estimate the incidence rate of DM for the years 2018-2023, an illness-death model was applied using a partial differential equation. This method has previously been used in the context of chronic non-communicable diseases[6,7]. This model is based on a mathematical relationship between the prevalence of DM, the mortality rate in the general population, and the MRR. Transitions between these states are characterized by incidence rates (no diabetes → diabetes) and mortality rates rates (no diabetes → death; diabetes → death).

The partial differential equation used to estimate the incidence of DM was developed by Brinks and Landwehr[6] and describes the change in prevalence with respect to age and calendar time:

$$i = \frac{\left(\frac{\partial}{\partial a} + \frac{\partial}{\partial t}\right) p}{1 - p} + m \cdot \frac{p(MRR - 1)}{p(MRR - 1) + 1} \tag{1}$$

Where i represents the incidence of DM, p represents the prevalence, and $\left(\frac{\partial}{\partial a} + \frac{\partial}{\partial t}\right) p$ denotes the temporal change in prevalence as a function of age a and calendar time t.Additionally, m reflects the mortality rate of the general population and MRR the mortality rate ratio. Therefore, the incidence rate can be calculated from estimates of the prevalence of DM, its temporal change by age and calendar time, the overall mortality rate, and the ratio of mortality rates between individuals with and without DM.

All statistical analyses will be performed in the R software version 4.5.0[8].

Results

During the study period, XXX participants from the ENDES population survey were included; of these, XXX (%) reported having been diagnosed with diabetes by their treating physician (Table 1). Figure 1 shows the prevalence of diabetes by age and sex, estimated from ENDES data using logit-transformed prevalence estimates. These estimates were used as input variables in the statistical model described above, along with the general population mortality rate and the MRR, to obtain the diabetes incidence rate for each year of analysis by sex (Figure 2) and the age-specific rate by sex for each year of analysis (Figure 3).

Between 2016 and 2023, the age-standardized incidence rate (IR) of diabetes showed a sustained increase in both sexes. Among men, the IR rose from 4.46 (95% CI: 3.61 to 5.23) in 2016 to 8.58 (95% CI: 5.27 to 14.41) in 2023, corresponding to an incidence rate ratio (IRR)

of 1.93 (95% CI: 1.14 to 3.24) compared with 2016. Similarly, among women, the IR increased from 5.20 (95% CI: 4.27 to 6.19) in 2016 to 9.87 (95% CI: 6.16 to 16.07) in 2023, with an IRR of 1.92 (95% CI: 1.15 to 3.16) relative to the reference year (Table 2).

When stratified by age group and calendar period, the IR increased between 2016 and 2023 across all age groups in both sexes. The oldest age groups exhibited the highest IRs in the population, reaching in 2023 an IR of 22.54 (95% CI: 13.49 to 36.57) among 60-year-old men and 25.76 (95% CI: 16.10 to 43.11) among 60-year-old women. However, analysis of the IRR indicated that the largest and statistically significant relative increases occurred in the younger age groups. Specifically, among 20-year-olds, the IRR was 2.18 (95% CI: 1.00 to 4.21) in men and 2.16 (95% CI: 1.03 to 4.28) in women, while the 30-year-old group showed a similar pattern, with IRRs of 2.13 (95% CI: 1.17 to 3.76) and 2.11 (95% CI: 1.25 to 3.52), respectively (Table 3).

Discussion

Understanding the epidemiological profile of DM in the Peruvian population is essential for informing public health strategies aimed at prevention and control[9]. There is a lack of cohort studies conducted in Latin American countries, which makes it unfeasible to directly estimate the incidence of DM. For this reason, this study will apply the illness-death model to estimate diabetes incidence using prevalence data from nationally representative surveys and mortality data from official vital statistics. This model has previously been implemented with Mexican data to calculate DM incidence rates for the period 2003-2015[5].

This study has some limitations. Diabetes status is defined by self-report and therefore may introduce information bias. Additionally, it is not possible to distinguish between type 1 and type 2 diabetes, which is important given the increasing incidence of type 1 diabetes reported in recent years[10]. On the other hand, a strength of this study is the use of nationally representative survey data from Peru. These findings will contribute recent estimates of diabetes incidence in Peru, which have not been previously reported and may inform public health planning and resource allocation.

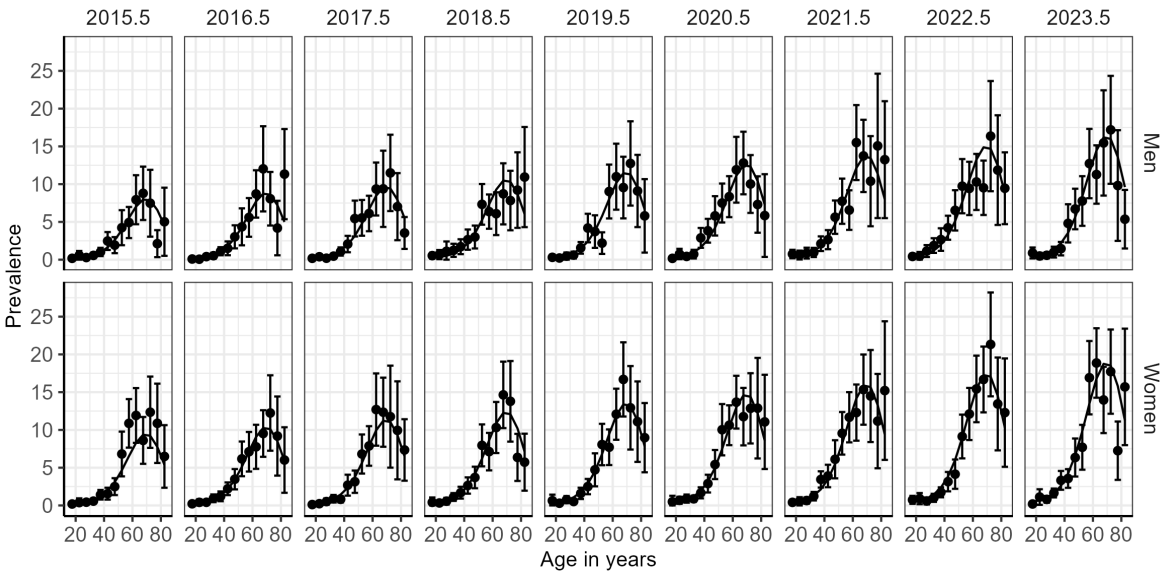
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Figures and Tables with legends

Table 1. Description of the Demographic and Family Health Surveys (ENDES) from Peru for the period 2018–2023


2015.5
2016.5
2017.5
2018.5
2019.5
2020.5
2021.5
2022.5
2023.5
Men
Women
Prevalence
0
5
10
15
20
25
20 40 60 80
Age in years

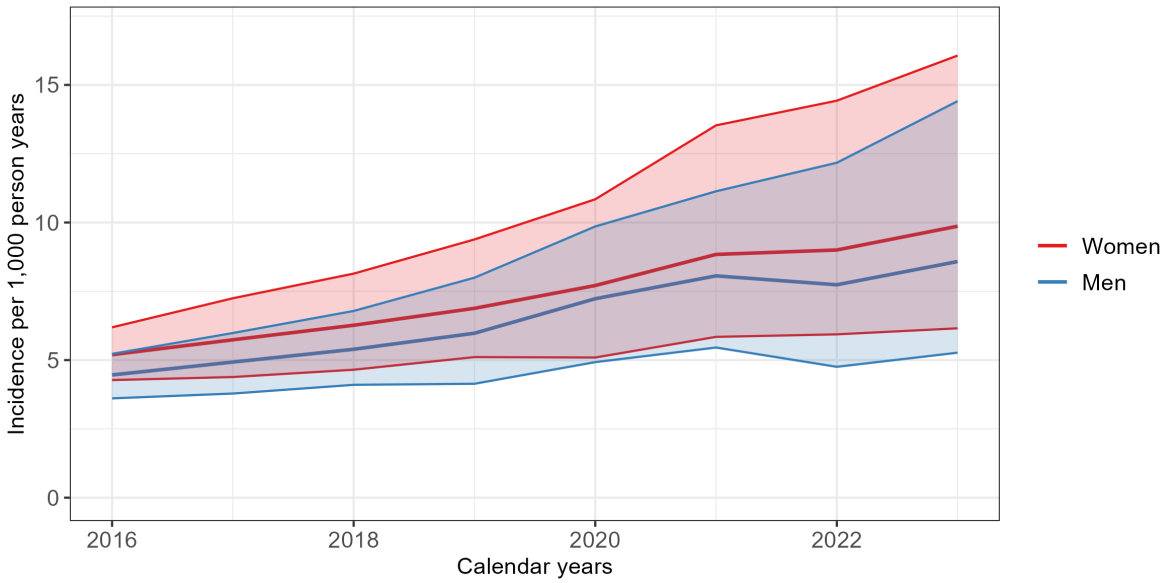

Incidence per 1,000 person years
0
5
10
15
2016
2018
2020
2022
Calendar years
Women
Men

Table 2. Age-standardized incidence rates of diabetes for men and women in Peru for the years 2016-2023, and the temporal change in incidence rates between 2016 and 2023

Year	Men – IR (95% CI)	Men – IRR (95% CI)	Women – IR (95% CI)	Women – IRR (95% C
2016	4.46 (3.61–5.23)	1.00 (1.00–1.00)	5.20 (4.27–6.19)	1.00 (1.00–1.00)
2017	4.93 (3.78–5.98)	1.10 (0.81–1.45)	5.74 (4.38–7.25)	1.11 (0.81–1.50)
2018	5.39 (4.10–6.79)	1.21 (0.90–1.64)	6.27 (4.65–8.14)	1.21 (0.87–1.65)
2019	5.98 (4.14–7.99)	1.34 (0.89–1.90)	6.88 (5.11–9.38)	1.34 (0.94–1.89)
2020	7.23 (4.93–9.86)	1.64 (1.08–2.33)	7.71 (5.09–10.84)	1.49 (0.97–2.22)
2021	8.06 (5.46–11.13)	1.81 (1.19–2.64)	8.84 (5.84–13.53)	1.71 (1.13–2.69)
2022	7.74 (4.76–12.17)	1.76 (1.07–2.79)	9.00 (5.94–14.43)	1.74 (1.13–2.89)
2023	8.58 (5.27–14.41)	1.93 (1.14–3.24)	9.87 (6.16–16.07)	1.92 (1.15–3.16)

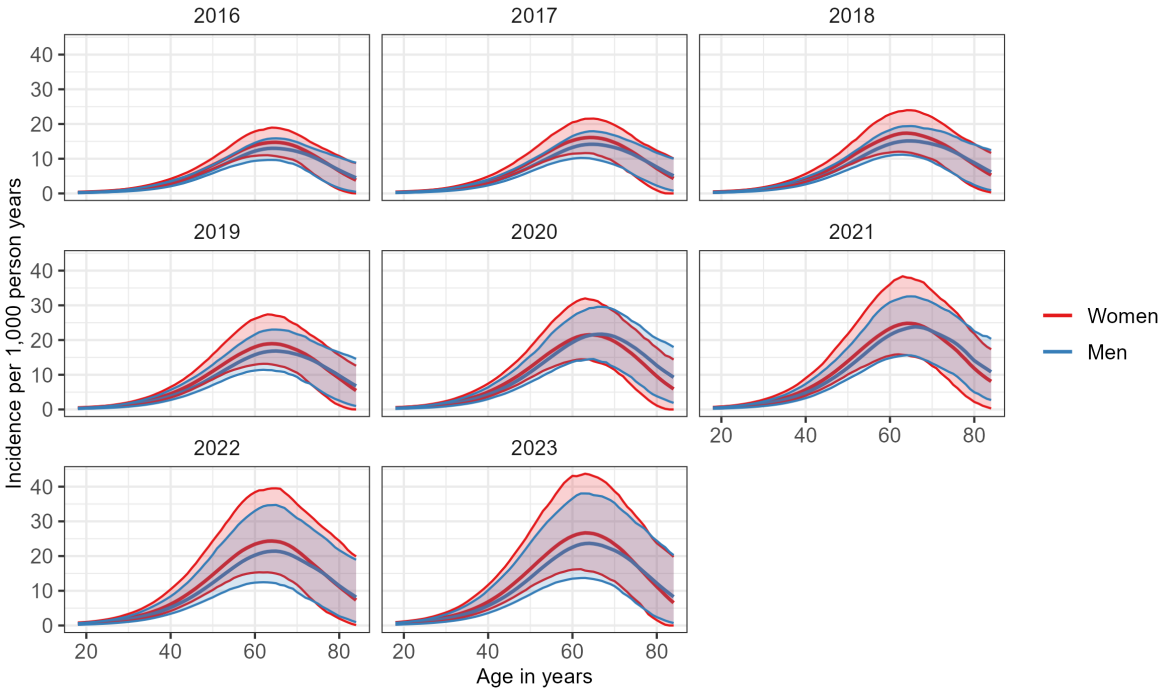

2016
2017
2018
2019
2020
2021
2022
2023
Incidence per 1,000 person years
Age in years
Women
Men

Table 3. Age and sex-specific incidence rates of diabetes in Peru for the years 2016-2023, and the temporal change in incidence rates between 2016 and 2023

age	Year	Men – IR (95% CI)	Men – IRR (95% CI)	Women – IR (95% CI)	Women – IF
20	2016	0.31 (0.18–0.40)	1.00 (1.00–1.00)	0.38 (0.21–0.47)	1.00 (1.00–
20	2023	0.68 (0.31–0.96)	2.18 (1.00–4.21)	0.83 (0.39–1.18)	2.16 (1.03–
30	2016	0.96 (0.62–1.17)	1.00 (1.00–1.00)	1.14 (0.84–1.39)	1.00 (1.00–
30	2023	2.00 (1.10–3.23)	2.13 (1.17–3.76)	2.39 (1.44–3.75)	2.11 (1.25–
40	2016	2.79 (2.06–3.55)	1.00 (1.00–1.00)	3.32 (2.76–4.20)	1.00 (1.00–
40	2023	5.71 (3.58–10.19)	2.06 (1.25–3.71)	6.83 (4.58–11.70)	2.07 (1.31–
50	2016	6.89 (5.82–8.23)	1.00 (1.00–1.00)	8.17 (7.07–10.18)	1.00 (1.00–
50	2023	13.74 (8.96–23.81)	1.99 (1.29–3.46)	16.26 (11.14–27.22)	1.99 (1.29–
60	2016	12.20 (9.39–14.60)	1.00 (1.00–1.00)	13.88 (10.83–17.87)	1.00 (1.00–
60	2023	22.54 (13.49–36.57)	1.86 (1.13–3.22)	25.76 (16.10–43.11)	1.87 (1.09–
70	2016	12.10 (8.08–14.79)	1.00 (1.00–1.00)	13.42 (9.20–17.05)	1.00 (1.00–
70	2023	21.63 (11.32–35.29)	1.80 (0.89–3.39)	23.39 (12.77–38.65)	1.77 (0.88–
80	2016	6.71 (1.55–10.41)	1.00 (1.00–1.00)	6.46 (1.57–10.79)	1.00 (1.00–
80	2023	12.17 (2.41–24.43)	1.86 (0.38–8.62)	11.18 (1.44–23.72)	1.74 (0.24–

References

1. Liu J, Ren ZH, Qiang H, et al. Trends in the incidence of diabetes mellitus: Results from the Global Burden of Disease Study 2017 and implications for diabetes mellitus prevention. *BMC Public Health*. 2020;20(1):1415. doi:10.1186/s12889-020-09502-x

2. Federation ID. *IDF Diabetes Atlas*. International Diabetes Federation; 2021. https://diabetesatlas.org

3. Seclen SN, Rosas ME, Arias AJ, Medina CA. Elevated incidence rates of diabetes in Peru: Report from PERUDIAB, a national urban population-based longitudinal study. *BMJ Open Diabetes Research & Care*. 2017;5(1):e000401. doi:10.1136/bmjdrc-2017-000401

4. Encuesta Demográfica y de Salud Familiar. ENDES. Published online 2024. https://proyectos.inei.gob.pe/endes/index.html

5. Silverman-Retana O, Brinks R, Hoyer A, Witte DR, Tönnies T. Using the illness-death model to estimate age- and sex-standardized incidence rates of diabetes in Mexico from 2003 to 2015. *BMC Public Health*. 2024;24(1):1882. doi:10.1186/s12889-024-19281-4

6. Brinks R, Landwehr S. Age- and time-dependent model of the prevalence of non-communicable diseases and application to dementia in Germany. *Theoretical Population Biology*. 2014;92:62-68. doi:10.1016/j.tpb.2013.11.006

7. Haß L, Tulka S, Tönnies T, et al. Age-specific incidence of need for long-term care for men and women in Germany 2015: Cross-sectional study comprising 82 million people. *F1000Research*. 2023;12:102. doi:10.12688/f1000research.129434.1

8. Team RC. R: A Language and Environment for Statistical Computing. Published online 2023. https://www.R-project.org/

9. Federation ID. How registries can help tackle the global diabetes burden. Published online 2024. https://idf.org/news/online-event-how-registries-can-help-tackle-the-global-diabetes-burden/

10. Gregory GA, Robinson TIG, Linklater SE, et al. Global incidence, prevalence, and mortality of type 1 diabetes in 2021 with projection to 2040: A modelling study. *The Lancet Diabetes & Endocrinology*. 2022;10(10):741-760. doi:10.1016/S2213-8587(22)00218-2